

Test 7 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. A bag holds 12 mangoes and 20 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

- A. 3:8
- B. 3:5
- C. 5:3
- D. 8:3

Show your work

multiple choice

2. Select **all** the numbers whose absolute value is 5.

Circle every correct choice:

- ☐ 5
- ☐ -5
- ☐ 10
- ☐ -10
- ☐ 0
- ☐ 6

Show your work

calculator · selectable hot text

3. Select **all** the points that lie on the vertical line through (5, 0).

- ☐ (7, 1)
- ☐ (5, -3)
- ☐ (5, 6)
- ☐ (2, 4)
- ☐ (5, 2)
- ☐ (7, -5)

Select all that apply.

Show your work

multiselect

4. Select **all** the expressions equivalent to 2 more than 5 times a number x .

- ☐ $7x$
- ☐ $5x + 2$
- ☐ $5(x + 2)$
- ☐ $2 + 5x$
- ☐ $x \cdot 5 + 2$
- ☐ $5x - 2$

Select all that apply.

Show your work

multiselect

5. Which fraction is equivalent to 60%, written in simplest form?

- A. $\frac{3}{5}$
- B. $\frac{6}{5}$
- C. $\frac{5}{3}$
- D. 6

Show your work

calculator · multiple choice

6. On a January morning the temperature in Homestead was 11 degrees below zero Celsius at a research freezer.

That temperature is $[[1]]$ °C, and its opposite is $[[2]]$ °C.

Blank (1) choices: -11 | 11 | -22 | 0

Blank (2) choices: -11 | 11 | -22 | 0

Show your work

calculator · editing task choice

7. The line plot shows books read last month.

What is the range of the data?



- A. 10
- B. 19
- C. 9
- D. 8

Show your work

multiple choice

8. A parallelogram has a base of 15 centimeters and a height of 5 centimeters.

Select **all** the expressions that give its area in square centimeters.

- 5×15
- 15×5
- $2 \times (15 + 5)$
- 75×1
- $(15 \times 5) \div 2$
- $15 + 5$

Select all that apply.

Show your work

multiselect

9. Select **all** the equations that have the solution $x = 37$.

☐ $x + 9 = 37$

☐ $x - 9 = 37$

☐ $x + 37 = 9$

☐ $9 + x = 46$

☐ $x + 9 = 46$

☐ $x - 9 = 28$

Select all that apply.

Show your work

multiselect

10. A bus travels 180 miles in 3 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

A. $\frac{1}{60}$ miles

B. 60 miles

C. 177 miles

D. 540 miles

Show your work

multiple choice

11. Marcos pays with \$20 for 5 items that cost \$8.01 each.

How much change should Marcos receive?

A. \$-12.04

B. \$11.99

C. \$40.05

D. \$-20.05

Show your work

calculator · multiple choice

12. Select **all** the expressions equivalent to $8(x + 9)$.

- ☐ $8(x + 10)$
- ☐ $72 + 8x$
- ☐ $8x + 9$
- ☐ $8x + 72$
- ☐ $17(x)$
- ☐ $8(x + 9)$

Select all that apply.

Show your work

multiselect

13. A right rectangular prism measures 8 by 7 by 2 inches.

Select **all** the expressions that give its volume in cubic inches.

- ☐ 8×7
- ☐ $2 \times 7 \times 8$
- ☐ $2 \times (8 + 7 + 2)$
- ☐ $8 + 7 + 2$
- ☐ 56×2
- ☐ $8 \times 7 \times 2$

Select all that apply.

Show your work

multiselect

14. A right rectangular prism measures 5 by 5 by 6 inches.

Select **all** the expressions that give its surface area in square inches.

- ☐ $5 \times 5 + 5 \times 6 + 5 \times 6$
- ☐ $2(5 \times 5 + 5 \times 6 + 5 \times 6)$
- ☐ $6 \times 5 \times 5$
- ☐ 170×1
- ☐ $5 \times 5 \times 6$
- ☐ $2(5 \times 5) + 2(5 \times 6) + 2(5 \times 6)$

Select all that apply.

Show your work

multiselect

15. Select **all** the points that are located in Quadrant III.

- (2, 3)
- (-1, -5)
- (-2, 6)
- (-5, -4)
- (-3, -2)
- (4, -1)

Select all that apply.

Show your work

multiselect

16. What is the value of $24 \div (-6)$?

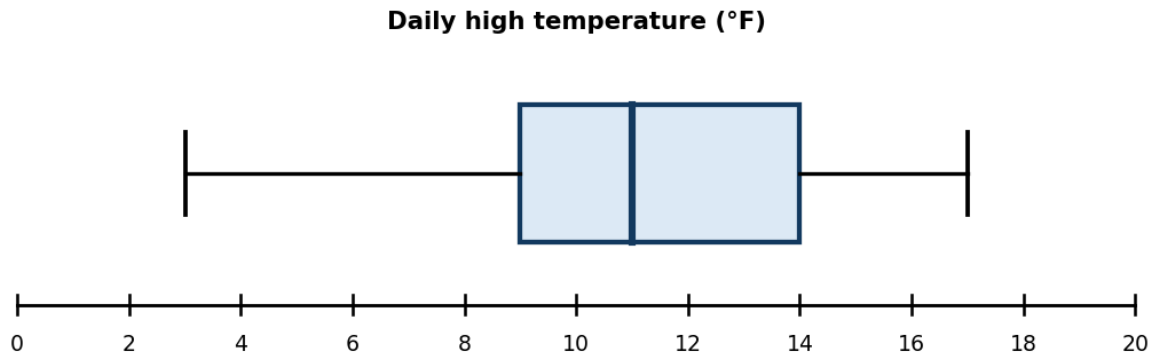
- A. 4
- B. 30
- C. -4
- D. -6

Show your work

calculator · multiple choice

17. The box plot shows a set of data.

What is the interquartile range of the data?



- A. 7
- B. 11
- C. 14
- D. 5

Show your work

multiple choice

18. The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

Boys	Girls	Total
3	5	8
9	15	?

Part-to-part-to-whole ratios

- A. 27
- B. 8
- C. 14
- D. 24

Show your work

multiple choice

19. A carry-on bag must weigh no more than 60 pounds.

Select **all** the values of x that make the situation true.

- ☐ 66
- ☐ 60
- ☐ 58
- ☐ 53
- ☐ 63
- ☐ 71

Select all that apply.

Show your work

multiselect

20. A data set is shown.

2, 4, 4, 5, 7, 9, 11, 13, 13, 14, 17, 18

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 10-14?

- A. 8
- B. 4
- C. 6
- D. 5

Show your work

multiple choice

21. A data set is shown.

7, 8, 19, 20, 21

The value 35 is added to the data set. What is the range of the new data set?

- A. 28
- B. 31
- C. 35
- D. 14

Show your work

multiple choice

22. 150 students is 75% of the students in a grade.

How many students are in the grade?

- A. 225
- B. 15000
- C. 11250
- D. 200

Show your work

multiple choice

23. What value of x makes the equation $\frac{2}{3} \times x = \frac{1}{3}$ true?

- A. $\frac{4}{3}$
- B. $\frac{2}{9}$
- C. 1
- D. $\frac{1}{2}$

Show your work

multiple choice

24. What is the value of $-3 - (-14)$?

- A. -17
- B. -11
- C. 11
- D. 14

Show your work

calculator · multiple choice

25. A roll holds 4 yards of ribbon. Isabella cuts 9 pieces that are each $\frac{3}{4}$ yard long.

How many yards of ribbon are left? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

26. A store sells avocados for \$2.70 each.

What is the total cost of 6 of them? Enter your answer in dollars.

Answer: _____

Show your work

calculator · equation editor

27. A number has the prime factorization 2×5^2 .

What is the number?

- A. 50
- B. 100
- C. 25
- D. 12

Show your work

calculator · multiple choice

28. Andres needs $4 \frac{1}{2}$ cups of flour for one batch of pastelitos.

How many cups are needed for 3 batches? Enter your answer as a fraction or a mixed number.

Answer: _____

Show your work

calculator · equation editor

29. Select **all** the values of x that make $x \leq 6$ true.

- ☐ 11
- ☐ 3
- ☐ 5
- ☐ 8
- ☐ 14
- ☐ 6

Select all that apply.

Show your work

multiselect

30. Store A sells 6 bottles of water for \$3.00. Store B sells 12 bottles for \$8.00.

What is the unit price at Store A, in dollars per bottle?

- A. 0.5
- B. 0.6667
- C. 9
- D. 2

Show your work

multiple choice

31. Select **all** the expressions with a value of 7 when $x = 2$.

- ☐ $2(x) + 3$
- ☐ $2(x + 3)$
- ☐ $2x + 3$
- ☐ $2x - 3$
- ☐ $3 + 2x$
- ☐ $5x$

Select all that apply.

Show your work

multiselect

32. Use the drop-down menus to place these numbers in order from least to greatest.

-5.5, $-1 \frac{3}{4}$, $\frac{1}{4}$, 3

Least , , , greatest.

Blank (1) choices: -5.5 | $-1 \frac{3}{4}$ | $\frac{1}{4}$ | 3

Blank (2) choices: -5.5 | $-1 \frac{3}{4}$ | $\frac{1}{4}$ | 3

Blank (3) choices: -5.5 | $-1 \frac{3}{4}$ | $\frac{1}{4}$ | 3

Blank (4) choices: -5.5 | $-1 \frac{3}{4}$ | $\frac{1}{4}$ | 3

Show your work

calculator · editing task choice

33. A triangle has a base of 14 inches and a height of 9 inches.

Select **all** the expressions that give its area in square inches.

- ☐ $(14 + 9) \div 2$
- ☐ 14×9
- ☐ $2 \times 14 \times 9$
- ☐ $0.5 \times 14 \times 9$
- ☐ $(14 \times 9) \div 2$
- ☐ $14 \times 9 \times 0.5$

Select all that apply.

Show your work

multiselect

34. What is the value of $(\frac{3}{4})^2$?

- A. $\frac{9}{16}$
- B. $\frac{3}{16}$
- C. $\frac{9}{4}$
- D. $\frac{3}{4}$

Show your work

calculator · multiple choice

35. Ximena recorded these scores:

3, 3, 5, 9, 10, 13, 14

What is the mode of the data set?

- A. 11
- B. 8
- C. 3
- D. 9

Show your work

multiple choice

36. What is the value of $9(3 + 1)$?

- A. 27
- B. 36
- C. 13
- D. 28

Show your work

calculator · multiple choice

37. Buses on one route leave Government Center every 9 minutes and buses on another route leave every 15 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

- A. 135
- B. 24
- C. 45
- D. 3

Show your work

calculator · multiple choice

38. Use the drop-down menu to make the statement true.

$$|-6| \text{ } \llbracket 1 \rrbracket \text{ } |16|$$

Blank (1) choices: < | > | =

Show your work

calculator · editing task choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume